

# TEXT MINING PROJECT - BBC NEWS ARTICLES

## CLASSIFICATION AND CLUSTERING OF NEWS ARTICLES

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## CHAPTER 1 INTRODUCTION AND PRE-PROCESSING

### 1.1 Introduction

This project focuses on the development of a complete text mining pipeline designed to support both classification and clustering tasks. Starting from raw textual data, the project investigates preprocessing strategies and text representation methods that are fundamental for building effective and reliable models.

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### 1.2 Objectives

The primary objectives of this project are:

- To design and implement a complete text mining pipeline for text classification and text clustering tasks.
  - To preprocess raw textual data in order to reduce noise and improve semantic consistency.
  - To apply different text representation techniques to transform documents into numerical features.
  - To compare the results obtained by classification and clustering models by distinguishing between different types of text representations and analyzing their impact on model performance.
  - To provide a reliable data foundation for downstream supervised and unsupervised learning approaches.
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### 1.3 Dataset

The dataset consists of BBC News articles belonging to five topical categories. This dataset consists of a collection of textual documents stored in a structured format. Each entry includes a text field representing the document content, along with additional attributes when available.

An initial exploratory data analysis was conducted to assess the dataset size, data quality, and text length distribution. This phase allowed the identification of missing values and inconsistencies, as well as an overall understanding of the corpus. The dataset was then prepared for preprocessing and feature extraction.

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## 1.4 Preprocessing

Text preprocessing is a crucial step for both text classification and text clustering, as raw textual data often contains noise and irrelevant variability. In this project, **two different preprocessing pipelines were implemented**, each tailored to the specific requirements of the adopted text representation techniques.

For frequency-based representations, such as Bag of Words and TF-IDF, the preprocessing pipeline focuses on strong text normalization. It includes conversion of text to lowercase, removal of punctuation, special characters, and numerical tokens, tokenization, stopword removal, and lemmatization. These steps aim to reduce vocabulary size and improve the discriminative power of textual features.

For embedding-based representations, a lighter preprocessing strategy was adopted in order to preserve semantic and contextual information. In this case, the text is minimally cleaned while avoiding aggressive normalization steps that could negatively affect the quality of pretrained language model embeddings.

This distinction allows a fair comparison between different representation types and ensures that each model operates on data processed according to its underlying assumptions.

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## 1.5 Data Representation

After preprocessing, textual documents were transformed into numerical representations suitable for machine learning models. Multiple text representation techniques were explored in order to support classification and clustering tasks.

Sparse representations such as Bag of Words and TF-IDF were employed to capture term frequency information and highlight discriminative terms across documents. These approaches are widely used and provide an effective baseline for text-based models.

In addition, dense vector representations based on pretrained language models were considered to capture semantic relationships between documents. These embeddings offer a more compact and semantically rich representation, which can be particularly beneficial when comparing classification performance and clustering structure across different representation types.

# CHAPTER 2 TEXT CLASSIFICATION

## 2.1 INTRODUCTION

This section presents a comparative analysis of different approaches for automatic classification of BBC news articles. The objective is to compare the effectiveness of different text representations (TF-IDF and BERT) combined with different machine learning algorithms (Logistic Regression, SVM and Random Forest).

The dataset was split into training and test sets using an 80/20 proportion, adopting a stratified sampling strategy to preserve the original class distribution. A fixed random seed (random\_state = 42) was used to ensure reproducibility of the experimental results.

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## 2.2 METHODOLOGY

### 2.2.1 Classification Algorithm

Three machine learning algorithms were employed for text classification, implemented using the TF-IDF and BERT text representations described in the previous section. These algorithms were chosen for their different characteristics and approaches to data modeling:

**Logistic Regression:** linear model optimized with max\_iter=1000 to ensure convergence.

**Support Vector Machine (SVM):** classifier with RBF kernel, effective for high-dimensional data.

**Random Forest:** ensemble of 100 decision trees, robust to noise and overfitting.

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### 2.2.2 Evaluation Metrics

Performance was evaluated using Accuracy (percentage of correct predictions) and Macro F1-Score (harmonic mean of precision and recall, calculated as unweighted average across classes).

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## 2.3 RESULTS AND DISCUSSION

### 2.3.1 Overall Performance

The following heatmap shows the performance of all embedding-model combinations in terms of Accuracy and Macro F1-Score:

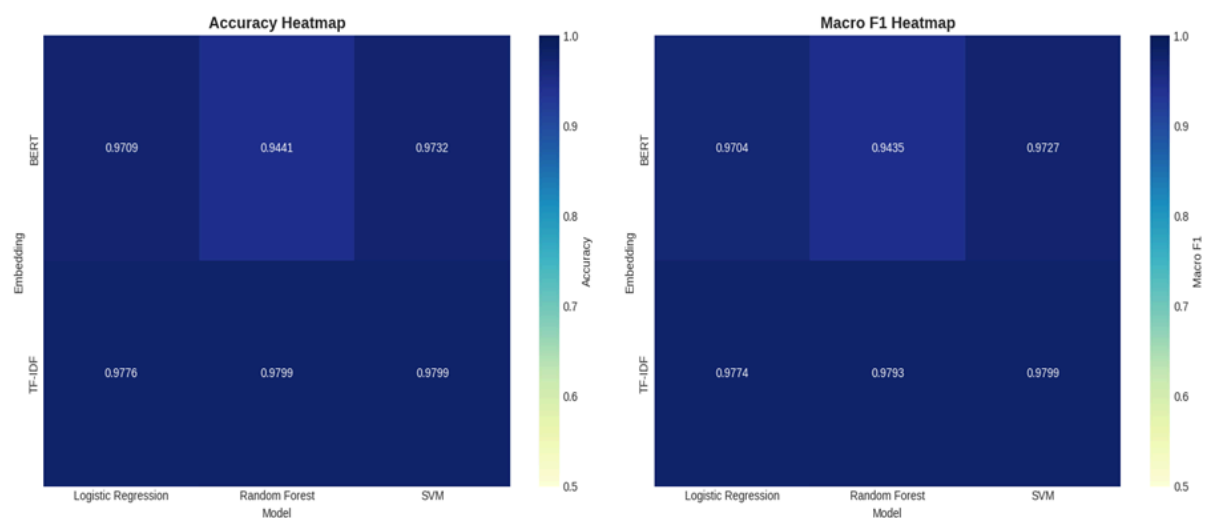


Figure 2.1: Performance heatmaps (Accuracy and Macro F1) for all configurations

| EMBEDDING | MODEL               | ACCURACY | MACRO F1 |
|-----------|---------------------|----------|----------|
| TF-IDF    | SVM                 | 0.9799   | 0.9799   |
| TF-IDF    | Random Forest       | 0.9799   | 0.9793   |
| TF-IDF    | Logistic Regression | 0.9776   | 0.9774   |
| BERT      | SVM                 | 0.9732   | 0.9727   |
| BERT      | Random Forest       | 0.9709   | 0.9704   |
| BERT      | Logistic Regression | 0.9441   | 0.9435   |

TAB 2.1: Results table for all configurations

The **TF-IDF + SVM** combination achieved the best performance with an Accuracy of 97.99% and a Macro F1-Score of 0.9799. The confusion matrix for the optimal configuration is shown below:

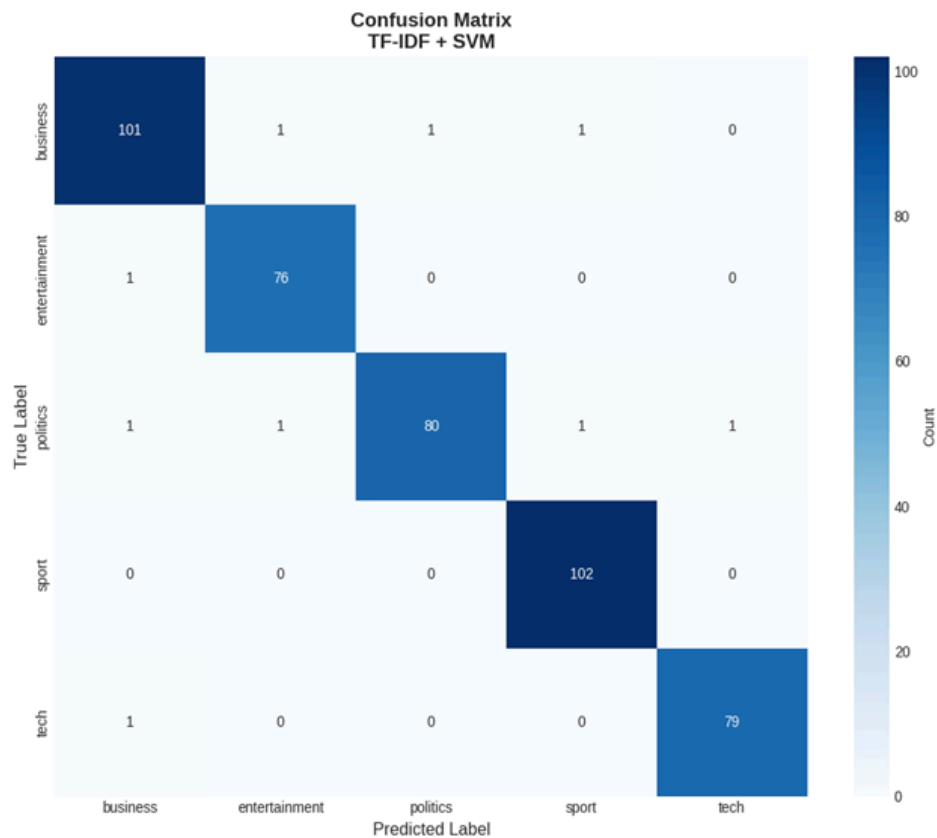


Figure 2.2: Confusion matrix for the best configuration (TF-IDF + SVM)

## 2.3.2 Classification Report

If we consider the results obtained with the best configuration (TF-IDF + SVM), the **Sport** category achieved the highest score with a perfect recall of 100%, while **Politics** shows the lowest recall (95.24%), indicating some difficulty in distinguishing political articles.

| CLASS         | PRECISION | RECALL | F1-SCORE | SUPPORT |
|---------------|-----------|--------|----------|---------|
| BUSINESS      | 0.9712    | 0.9712 | 0.9712   | 104     |
| ENTERTAINMENT | 0.9744    | 0.9870 | 0.9806   | 77      |
| POLITICS      | 0.9877    | 0.9524 | 0.9697   | 84      |
| SPORT         | 0.9808    | 1.0000 | 0.9903   | 102     |
| TECH          | 0.9875    | 0.9875 | 0.9875   | 80      |

TAB 2.2: Results Table for each class

The following figure presents the confusion matrices for all six embedding-model combinations tested:

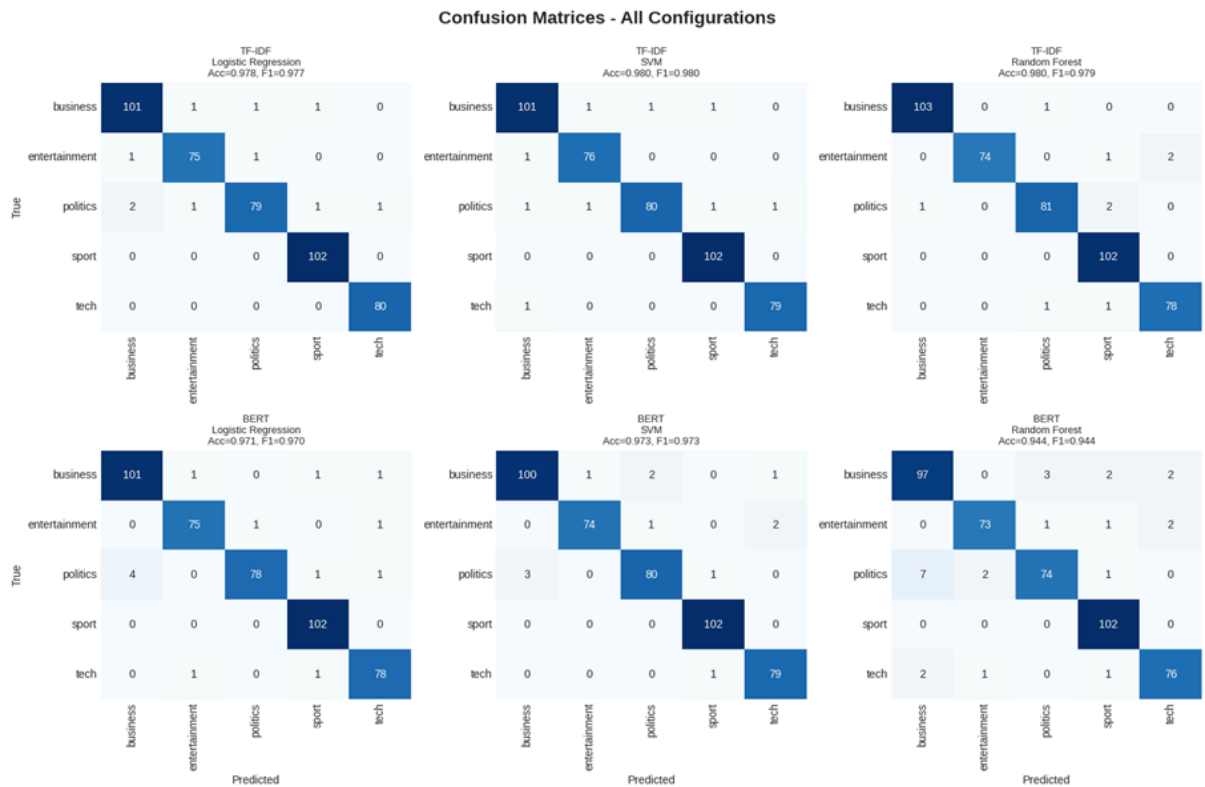


Figure 2.3: Confusion matrices for all embedding-model combinations

### 2.3.3 Discussion

Contrary to expectations, the TF-IDF representation outperformed BERT embeddings in all configurations. This result can be attributed to several factors: the nature of the task (news classification with distinctive vocabulary per category), the relatively small dataset size, and the fact that TF-IDF effectively captures the key terms that characterize each news category.

SVM and Random Forest showed similar performance with TF-IDF, while Random Forest had a significant drop with BERT (94.41% vs 97.32% for SVM). This suggests that Random Forest is more sensitive to the quality of text representation, while SVM maintains more stable performance.

From the confusion matrices, it emerges that the most common errors involve confusion between Business and Politics, categories that often share economic and political terminology. The Sport category is the most easily identifiable, likely due to its highly specialized vocabulary.

## 2.3.4 Conclusions

The results highlight a clear performance difference between traditional TF-IDF-based representations and contextual embeddings derived from BERT across all evaluated classifiers. In particular, the combination of TF-IDF features and Support Vector Machines achieves the highest Macro F1 score, outperforming both Logistic Regression and Random Forest models, as well as all configurations based on BERT embeddings.

This outcome, which may appear counterintuitive given the effectiveness of transformer-based models, can be explained by the nature of the task and the adopted experimental setup. The BBC dataset consists of well-defined topical categories characterized by distinctive vocabularies, a scenario in which sparse lexical representations such as TF-IDF prove to be particularly effective.

The analysis of the confusion matrix for the best-performing model shows that most misclassifications occur between semantically related classes, particularly Business and Politics, which frequently share similar terminology in news articles. In contrast, categories such as Sport and Tech exhibit higher classification accuracy, likely due to the presence of more specific and less ambiguous lexical patterns.

# CHAPTER 3 TEXT CLUSTERING

## 3.1 INTRODUCTION

Unsupervised text clustering was performed to analyze the intrinsic structure of the BBC News corpus and to assess how well different text representations capture semantic similarities between documents. Both sparse and dense representations were considered, and multiple clustering algorithms were evaluated.

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## 3.2 METODOLOGY

### 3.2.1 Clustering Setup

Two types of document representations were used:

- **TF-IDF vectors**, providing high-dimensional sparse representations based on term frequency statistics.
- **Sentence embeddings**, obtained from a pretrained transformer model and normalized to support cosine similarity.

Clustering was performed using:

- **KMeans**, which partitions documents by minimizing within-cluster variance.
- **Agglomerative Clustering**, which builds hierarchical clusters using different linkage criteria.

The number of clusters was selected empirically through **silhouette analysis**, evaluating values of (  $k$  ) between 2 and 20 using cosine distance.

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### 3.2.2 Selection of the Number of Clusters

Silhouette score analysis highlighted different optimal cluster configurations depending on the representation:

- For **TF-IDF + KMeans**, the best silhouette score was achieved with  **$k = 16$** , suggesting a finer-grained partition of the corpus when using sparse representations.
- For **Embeddings + KMeans**, the silhouette score peaked at  **$k = 5$** , which aligns with the known number of BBC News categories.

These results indicate that dense embeddings better capture high-level semantic groupings, while TF-IDF representations tend to emphasize lexical variability.

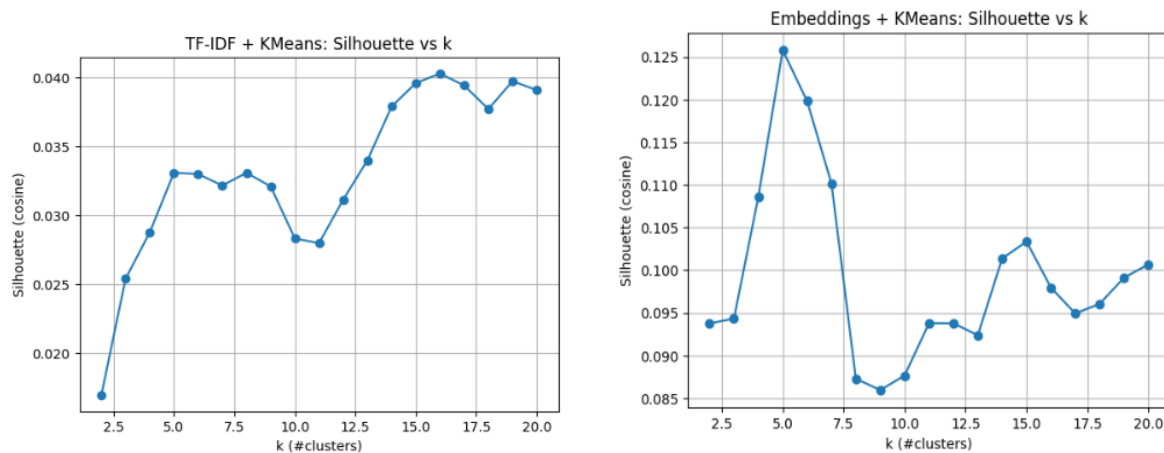


Fig 3.1: Silhouette score as a function of the number of clusters ( $k$ ) for TF-IDF and embedding-based representations using KMeans clustering.

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### 3.2.3 Clustering Evaluation Metrics

Clustering quality was evaluated using both **internal** and **external** metrics:

- **Silhouette Score** (internal), measuring cluster compactness and separation.
- **Adjusted Rand Index (ARI)**, **Normalized Mutual Information (NMI)**, and **V-measure**, comparing predicted clusters with ground truth labels.
- **Purity**, assessing the dominant class proportion within each cluster.
- **Macro F1-score (mapped)**, obtained by mapping each cluster to its most frequent true label.

Among all tested configurations, **Embeddings + KMeans ( $k = 5$ )** achieved the best overall performance, with high ARI, NMI, purity, and macro F1-score. This confirms the effectiveness of contextual embeddings for capturing semantic similarities between documents.

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## 3.3 RESULTS AND DISCUSSION

### 3.3.1 Cluster Visualization

To better interpret the clustering results, a **Principal Component Analysis (PCA)** was applied to reduce the embedding space to two dimensions. Visual inspection shows that the clusters obtained with embeddings are well separated and correspond closely to the true BBC categories.

Additional comparison plots between predicted clusters and ground truth labels further confirm a strong alignment between unsupervised clustering results and the original class structure.

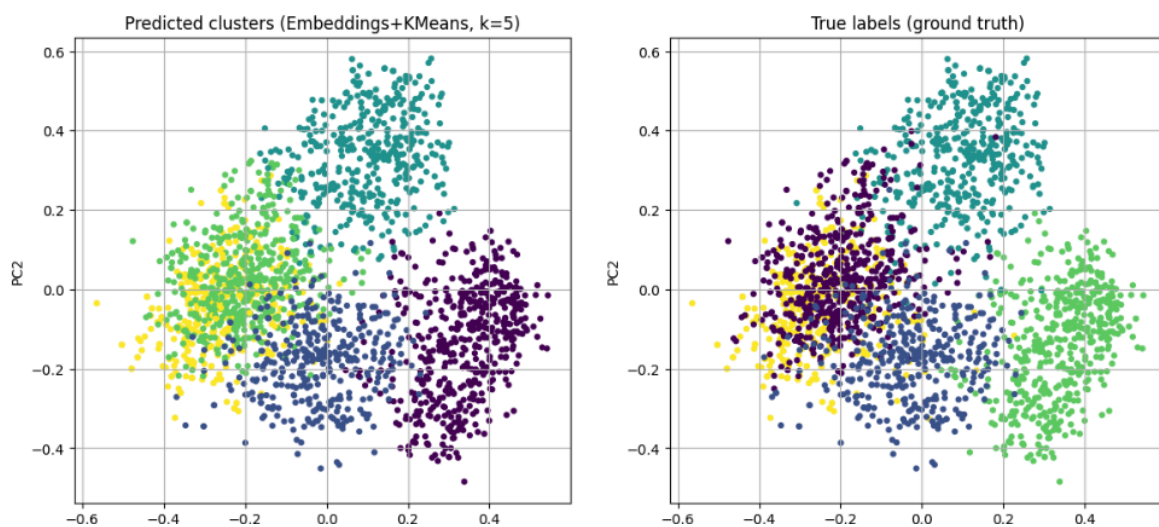


Fig 3.2: PCA-based visualization of document embeddings. Left: clusters predicted by KMeans ( $k=5$ ). Right: true BBC News labels.

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### 3.3.2 Cluster Interpretation and Analysis

A detailed analysis of cluster composition was conducted:

- **Cluster size distribution** revealed a balanced allocation of documents across clusters, indicating the absence of dominant or degenerate clusters.
- **Cross-tabulation between true labels and predicted clusters** showed that most clusters are highly homogeneous, with a clear dominant category.
- For TF-IDF-based clustering, **top terms per cluster** were extracted from centroid weights, providing interpretable keywords that characterize each group and align well with the corresponding news topics.

These analyses improve the interpretability of clustering results and validate the semantic coherence of the discovered clusters.

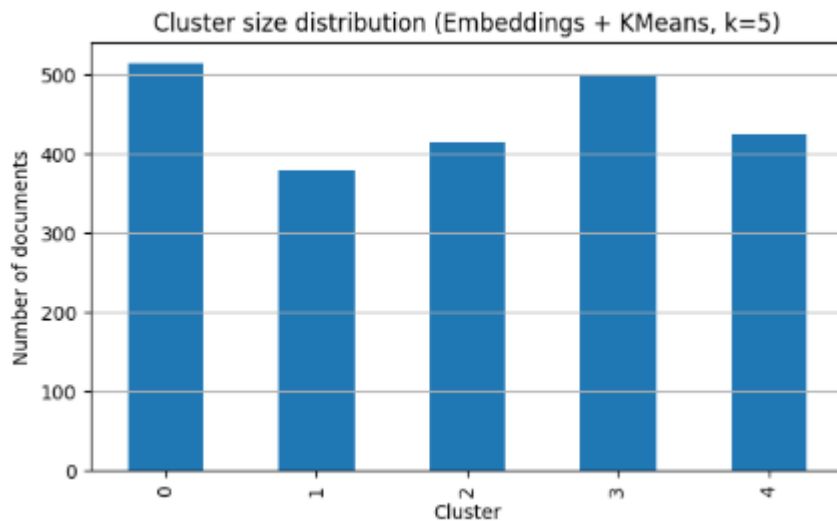


Fig 3.3: Distribution of documents across clusters for the embedding-based KMeans solution

CROSSTAB: True labels vs Predicted clusters (Embeddings + KMeans)

| Cluster (k=5) | 0   | 1   | 2   | 3   | 4   |
|---------------|-----|-----|-----|-----|-----|
| True label    |     |     |     |     |     |
| business      | 2   | 1   | 13  | 483 | 18  |
| entertainment | 0   | 366 | 3   | 3   | 14  |
| politics      | 3   | 1   | 397 | 12  | 4   |
| sport         | 507 | 3   | 1   | 0   | 0   |
| tech          | 3   | 8   | 0   | 2   | 388 |

Fig 3.4: Cross-tabulation between true BBC categories and predicted clusters (row-normalized percentages).

- Cluster 0:** [ club, chelsea, game, arsenal, league, play, united, liverpool, goal, player, manager, win ]
- Cluster 1:** [ bn, share, firm, company, profit, sale, market, year, airline, deal, stock, us]
- Cluster 2:** [ chart, number, elvis, top, single, box, album, presley, office, film, download, release ]
- Cluster 3:** [ yukos, oil, russian, gazprom, russia, yugansk, rosneft, crude, price, company, bn, barrel ]
- Cluster 4:** [ olympic, race, indoor, win, athens, athlete, world, iaaf, champion, kenteris, championship, medal ]
- Cluster 5:** [ ukip, kilroy, silk, party, mr, election, donation, houston, verita, candidate, mep, hockney ]
- Cluster 6:** [ mr, labour, election, blair, party, tory, brown, howard, would, tax, minister, chancellor ]
- Cluster 7:** [ mr, lord, law, eu, court, government, would, police, case, minister, right, legal ]
- Cluster 15:** [england, wale, rugby, ireland, game, nation, france, robinson, six, coach, side, win ]

Top TF-IDF terms characterizing each cluster.

### 3.3.3 Discussion

The clustering experiments demonstrate that:

- Dense embedding representations significantly outperform sparse TF-IDF vectors in capturing semantic document similarity.
- KMeans combined with sentence embeddings is particularly effective when the expected number of clusters is known.
- External evaluation metrics confirm that unsupervised clustering can recover meaningful class structures without using label information during training.

Overall, the clustering results complement the classification experiments and provide additional insights into the latent organization of the BBC News corpus.

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## CHAPTER 4 FINAL REMARKS

Overall, the experimental results highlight the impact of text representation choices on both classification and clustering performance. In the supervised setting, traditional TF-IDF representations combined with linear classifiers, particularly Support Vector Machines, achieved the best results, outperforming contextual embeddings due to the distinctive lexical characteristics of the news categories. Conversely, in the unsupervised setting, dense embedding representations proved more effective in capturing high-level semantic similarities between documents, leading to well-separated and semantically coherent clusters that closely align with the true category structure. These findings demonstrate that no single representation is universally optimal, and that the effectiveness of text mining models strongly depends on the task and the underlying representation strategy.